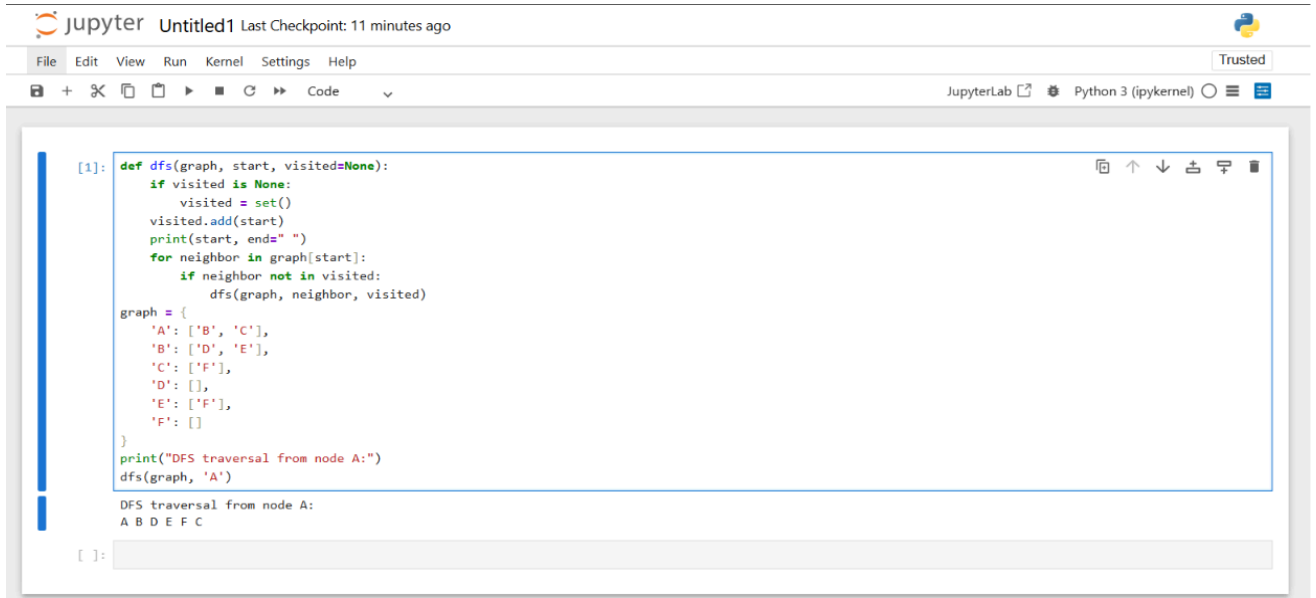


Depth First Search (DFS)

Python Code:

```
def dfs(graph, start, visited=None):
    if visited is None:
        visited = set()
    visited.add(start)
    print(start, end=" ")
    for neighbor in graph[start]:
        if neighbor not in visited:
            dfs(graph, neighbor, visited)
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}
print("DFS traversal from node A:")
dfs(graph, 'A')
```

Output:



The image shows a JupyterLab interface with a code editor and an output area. The code editor contains a Python function `dfs` that performs a depth-first search on a graph. The graph is defined as a dictionary where keys are nodes and values are lists of neighbors. The function starts at node 'A' and traverses the graph, printing the nodes in the order they are visited. The output area shows the result of the traversal: "DFS traversal from node A:" followed by the sequence "A B D E F C".

```
[1]: def dfs(graph, start, visited=None):  
    if visited is None:  
        visited = set()  
    visited.add(start)  
    print(start, end=" ")  
    for neighbor in graph[start]:  
        if neighbor not in visited:  
            dfs(graph, neighbor, visited)  
  
    graph = {  
        'A': ['B', 'C'],  
        'B': ['D', 'E'],  
        'C': ['F'],  
        'D': [],  
        'E': ['F'],  
        'F': []  
    }  
    print("DFS traversal from node A:")  
    dfs(graph, 'A')
```

DFS traversal from node A:
A B D E F C